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The statistical reconciliation of time series of accounts between two benchmark revisions

Baoline Chen[1] | Tommaso Di Fonzo[2] | Thomas Howells[1] | Marco Marini[3]

[1]Bureau of Economic Analysis, Washington DC, USA

[2]Department of Statistical Sciences, University of Padova, Padova, Italy

[3]Statistics Department, International Monetary Fund, Washington DC, USA

Correspondence
Tommaso Di Fonzo, Department of Statistical Sciences, University of Padova, 35121 Padova, Italy.
Email: difonzo@stat.unipd.it

The 2003–2007 U.S. annual input–output accounts, GDP-by-industry accounts, and expenditure-based GDP are reconciled with the 2002 and 2007 quinquennial benchmarks and all contemporaneous constraints of the input–output accounts for the in-between years. The reconciliation is performed at a very detailed level (six-digit NAICS) according to feasible statistical procedures able to deal with very large systems of accounts. Our objective is to adjust the preliminary levels of the annual 2003–2007 series such that they are consistent with the quinquennial benchmarks available, fulfill all the accounting relationships for any given year, and show movements that are as close as possible to the preliminary information. We use a simultaneous least-squares procedure based on the proportional first difference criterion, a well known movement preservation principle proposed by Denton. We evaluate the possible adoption of (i) a pure proportional (PROP) adjustment for small series and series with both negative and positive values that deteriorate the meaningfulness of growth rates, and (ii) a priori assumptions for groups of variables according to their different reliabilities, where this can reasonably be imposed.

KEYWORDS

alterability coefficients, benchmarking, input–output accounts, reconciliation, temporal and contemporaneous constraints

1 | INTRODUCTION

Measurements of socioeconomic phenomena are conducted at different frequencies and with different objectives. Monthly or quarterly information aims at providing a timely picture of the short-term movements. Annual data from sample surveys or administrative statistics from regulatory agencies rely on a large sample of units, and thus, they provide a more accurate indication of medium- and long-term trends than intra-annual data. The Economic Census collects the most comprehensive data available on business activities and provides a detailed and accurate portrait of the country's economy once every five years (e.g., for the U.S.A.) or more.

Theoretically, higher frequency measurements should be consistent with lower frequency benchmarks. Of course, this is rarely the case in practice, and consistency must be imposed on the data. Furthermore, at each frequency, social-economic variables may be required to satisfy a number of aggregation and accounting relationships. A typical example is national accounts, where total aggregates of the economy must be consistent with the sum of detailed components (e.g., by industry or by commodity) and identities are established between flows of production, expenditure, and income. However, cross-sectional consistency among the observed variables is not automatically met and must be imposed on the data.

In a time series of a system of accounts, observed data need to be adjusted such that both temporal and contemporaneous constraints are satisfied. A reconciliation process aims at preserving as much as possible the content of the preliminary information available. Because the time series dimension of socioeconomic variables is relevant, it is often requested that the temporal dynamics (i.e., the growth rates) of the preliminary information be preserved in the best possible way.

The standard mathematical/statistical procedures used to balance the accounts in each time period do not take into account the time dimension, which does not play any role when classical techniques like biproportional matrix balancing procedure RAS (Bacharach, 1970) or the least-squares approach by Stone, Champernowne, and Meade (1942) are applied to the preliminary accounts of successive time periods. However, having a reliable, comprehensive, and consistent dynamic view of the economy as described by a time series of accounts is very important for most users. On this theme, in the last three decades, several contributions combining the least-squares approach by Stone et al. (1942) and the seminal paper by Denton (1971) on temporal benchmarking of a time series have been proposed in the literature (Dagum & Cholette, 2006; Di Fonzo, 1990; Di Fonzo & Marini, 2005, 2006; Quenneville & Fortier, 2012; Sefton & Weale, 1995; Solomou & Weale, 1993, 1996; van der Ploeg, 1982). The consideration of the temporal dimension in the RAS balancing framework seems less immediate. For a recent proposal to disaggregate input-out tables in time, see the work of Avelino (2017). In addition, the computational burden, which originally represented an obstacle to the development of reconciliation procedures for medium-large systems, seems now to be less important (Di Fonzo & Marini, 2011, 2015), which makes this framework readily applicable even in many real-life situations faced by national accountants.

In a recent study (Chen, Di Fonzo, & Marini, 2013), a specific problem of reconciling annual (preliminary) estimates of U.S. national accounts aggregates subject to quinquennial benchmarks available from input–output (IO) accounts at summary level of detail[1] was addressed. Given preliminary, revised, but not fully balanced annual IO accounts from 1998 to 2002 and two revised and fully balanced IO accounts for benchmark years of 1997 and 2002, annual IO accounts for the years 1998–2001 were fully balanced and revised, with the temporal profile of the preliminary aggregates preserved as much as possible. The objectives were to adjust the annual data such that they (i) were consistent with the quinquennial benchmarks available, (ii) fulfill all the

[1]Estimates in the Industry Economic Accounts of the Bureau of Economic Analysis (BEA) have been generally available at three levels of detail according to the 2007 North American Industry Classification System (NAICS) code structure: sector (15 industry groups), summary (71 industry groups), and detail (389 industry groups). For most data products, published estimates at the detail level are available only for estimate year 2007 (due to the availability of detailed data from the 2007 Economic Census). See the work of Young, Howells, Strassner, and Wasshausen (2015).

IO accounting relationships for any given year, and (iii) show movements that are as close as possible to the preliminary information. A simultaneous least-squares procedure based on the proportional first difference (PFD) criterion, a movement preservation principle proposed by Denton (1971) and modified by Cholette (1984) to correctly deal with the starting condition, was compared with a pure proportional (PROP) adjustment procedure.

The results showed that these objectives were best achieved through the least-squares procedure based on the PFD criterion because this procedure was able to smooth the differences observed between the preliminary and the benchmark data, reducing the impact of the correction by distributing it over all the years. However, it was also noticed that a PFD adjustment provides unsatisfactory results for a small subset of very small series and series presenting changes from positive to negative values. Because these movements are difficult to preserve, they were adjusted according to a PROP criterion. It was shown that a constrained optimization procedure that minimizes a combined PFD–PROP objective function improves the overall adjustment of the system, minimizing the impact on the year-to-year changes of the preliminary series. PROP is not the only procedure to handle the series with nonmeaningful temporal movements. The additive first difference procedure by Denton (1971) is another approach that has been attempted (Bikker, Daalmans, & Mushkudiani, 2013; Chen, 2007). A merit of the PFD–PROP reconciliation procedure is that it has the property of invariance of input data (i.e., scale invariance): "If all input data are multiplied by the same nonnegative scalar, the outcomes must also be changed by this factor" (Bikker et al. 2013, p. 397).

The release of the 2007 benchmark IO tables in January 2014 made it possible to perform a similar study, that is, at the summary level of detail, for the 2003–2007 period (Chen, Di Fonzo, Howells, & Marini, 2014). To demonstrate the feasibility of this simultaneous procedure at close to the actual production level of detail, in this paper, we extend this latest study to work on a significantly more disaggregated version of the IO accounts (six-digit NAICS), which consists of a total of 159,952 nonzero annual time series. We show that our simultaneous constrained optimization procedure to reconcile the preliminary unbalanced annual IO tables for 2003 through 2007 with the fully balanced 2002 and 2007 benchmark IO tables at a very detailed level is able to produce revised, fully balanced, and very detailed IO accounts for the years 2003–2006, where the temporal profile of the preliminary aggregates is preserved as much as possible. In addition, the adopted statistical framework permits to consider the reliability of the aggregates, to limit (or avoid at all) the adjustment to some variables in the system (Dagum & Cholette, 2006; Stone et al., 1942; van der Ploeg, 1982). This can be done in a straightforward way, giving the user (i.e., national accountants) a powerful tool to manage this delicate phase of the accounts production.

This paper is structured as follows. Section 2 describes the construction of the U.S. benchmark IO tables and the revision of previously published annual IO estimates. Section 3 briefly introduces benchmarking and reconciliation of economic time series. Section 4 presents and evaluates the results achieved using a least-squares procedure based on alternative objective functions. Section 5 considers the use of alterability coefficients to take into account different reliabilities of the variables involved in the reconciliation. Finally, Section 6 concludes this paper.

2 | CONSTRUCTION AND REVISION OF U.S. INPUT-OUTPUT ACCOUNTS

The U.S. national accounts system measures gross domestic product (GDP) via the production, expenditure, and income approaches. For the system to be consistent, production-based GDP measured as total output less total inputs from the IO accounts, expenditure-based GDP measured as total final expenditures from the national income and product accounts (NIPAs), and gross domestic income as measured in the NIPAs must all be reconciled.

The U.S. IO accounts consist of make and use tables that are classified into N industries and M commodities. This implies that, at a minimum, the IO accounts must satisfy N sets of industry and M sets of commodity cross-sectional or contemporaneous accounting constraints for each period. Because data used to compile the IO accounts are obtained from a variety of sources, inconsistency often arises in the initial estimates due to differences in the definition and classification of source data items and to measurement errors in the source data. Consequently, initial estimates rarely satisfy all contemporaneous accounting constraints, and some technique must be employed to impose consistency among the components of the system.

In addition to the contemporaneous constraints, each component series in the annual IO accounts must be consistent with the corresponding quinquennial benchmark levels, and at the same time, the individual series are required to show a realistic temporal profile in the years between the benchmarks as well. The benchmark IO tables, based primarily on data from the Economic Census, contain more complete information and are typically more accurate than higher frequency series; however, these tables are far less timely as they are published only every five years and with a significant lag. Higher frequency source data used to estimate quarterly GDP-by-industry or annual IO tables are more timely, but they often contain incomplete information and are therefore less accurate. As a result, benchmarking or interpolation procedures must also be employed to align less accurate, higher frequency data with less frequent, higher accuracy data. A sensible way to achieve internal consistency within each period while minimizing the disruption to period-to-period movements in each series is to impose contemporaneous and temporal aggregation constraints simultaneously. However, the method employed to prepare the published U.S. IO tables is a sequential procedure.

To begin with, the 2007 benchmark make table was constructed primarily using data from the 2007 Economic Census (Young et al., 2015). Estimates in the make table were considered predetermined and were not adjusted during the balancing of the benchmark IO tables. Next, an unbalanced 2007 benchmark use table was constructed using data from a variety of sources, including the Census Bureau, the Internal Revenue Services, the Bureau of Labor Statistics, and other public and private data sources. Initial value added (VA) estimates in the unbalanced table were based on production-based and income-based VA estimates that were reconciled using a weighted least-squares minimization technique. Final balancing of the use table was conducted using the RAS procedure, subject to predetermined marginal aggregates that included gross output by industry and by commodity, final uses by category and by commodity, and reconciled VA estimates.

Following the U.S. national accounts' sequential revision procedure, annual make table estimates from 2003 to 2006 were converted from the 2002 to the 2007 NAICS classification system and then benchmarked to the 2007 benchmark make table using the Denton PFD procedure. Annual use tables were also converted from 2002 to 2007 NAICS and updated to incorporate information from the revised NIPAs. Tables for each period were then balanced independently using RAS, subject to predetermined marginal aggregates similar to those used in the final balancing of the benchmark use table. For more details on the integration of the benchmark and annual IO accounts, see the works of Strassner and Wasshausen (2013) and Kim, Strassner, and Wasshausen (2014).

It is well known that, as a numerical procedure for balancing accounts in each period, RAS does not allow varying degrees of uncertainty in initial estimates and constraints, provides

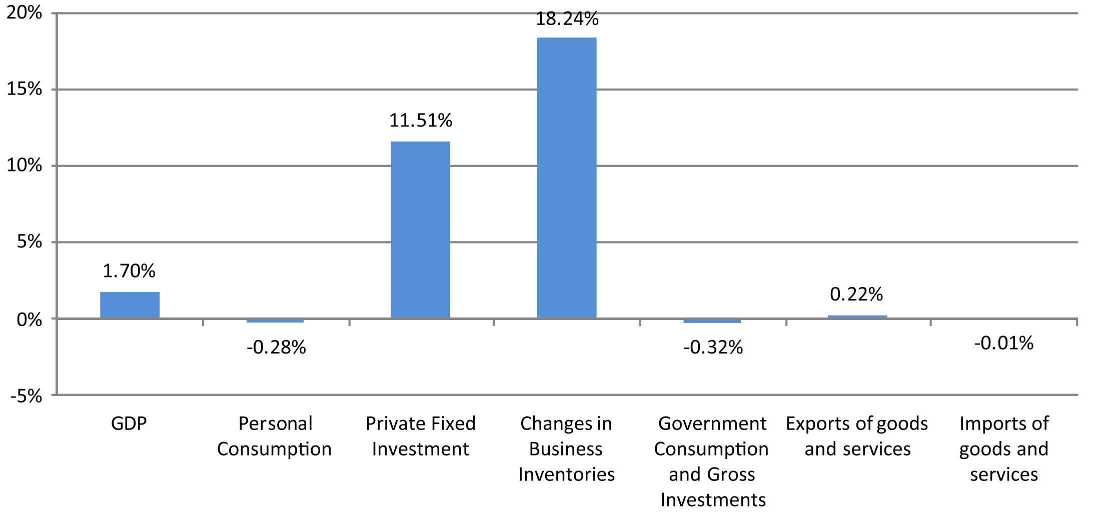

FIGURE 1 2007 benchmark revision of gross domestic product (GDP) and final uses by major category (percentage of preliminary 2007 values)

dubious economic interpretation of the balancing results, and requires many iterations for convergence (Dagum & Cholette, 2006). An added disadvantage of using the RAS procedure to reconcile time series accounts is that it does not consider the temporal profile of the component series in the accounts.

Note that, when a benchmark revision occurs, both the levels and growth rates of the variables in the IO tables are affected. Figure 1 shows the percentage revision produced by the 2007 benchmark estimates on the level of GDP and the major final uses aggregates. The impact of the 2007 benchmark revision ranges from −0.32% for government consumption and gross investments to 18.24% for changes in business inventories, whereas the 2007 preliminary GDP level showed a 1.70% upward revision. All these differences between preliminary and final (benchmark) data can be seen as temporal inconsistencies, which require the adjustment of annual figures between the two benchmark periods.

The preliminary annual IO tables were also unbalanced. Considering a level of detail of 65 industries (three-digit NAICS) and 67 commodities, the series present temporal inconsistencies and accounting discrepancies by industry and by commodity. Figures 2 and 3 show the 10 industries and 10 commodities with the largest discrepancies. Unlike the other industries and commodities, discrepancies for these categories are large in 2003–2005, although they are close to zero in 2006.

In this paper, we extend on (Chen et al., 2014) working on a significantly more disaggregated version of the IO accounts (six-digit NAICS), which considers 830 industries, 852 commodities, 455 categories of final expenditures, and three categories of VA. At this level of detail, the system of IO accounts consists of a total of 1,804,470 series, 707,160 from the make table, and 1,097,310 from the use table. Of the 707,160 from the make table, 9,069 are nonzero series, and of the 1,097,310 series from the use table, the nonzero series include 143,896 intermediate inputs, 2,406 values added, and 4,221 final expenditures. As shown in Table 1, the number of nonzero series changes from 4,842 at the three-digit NAICS level of detail to 159,592 at the six-digit NAICS level of detail (the percentage of nonnull series decreases from 50.22% to 8.84%), which clearly amounts to a very large reconciliation problem, as compared with that of the work of Chen et al. (2014).

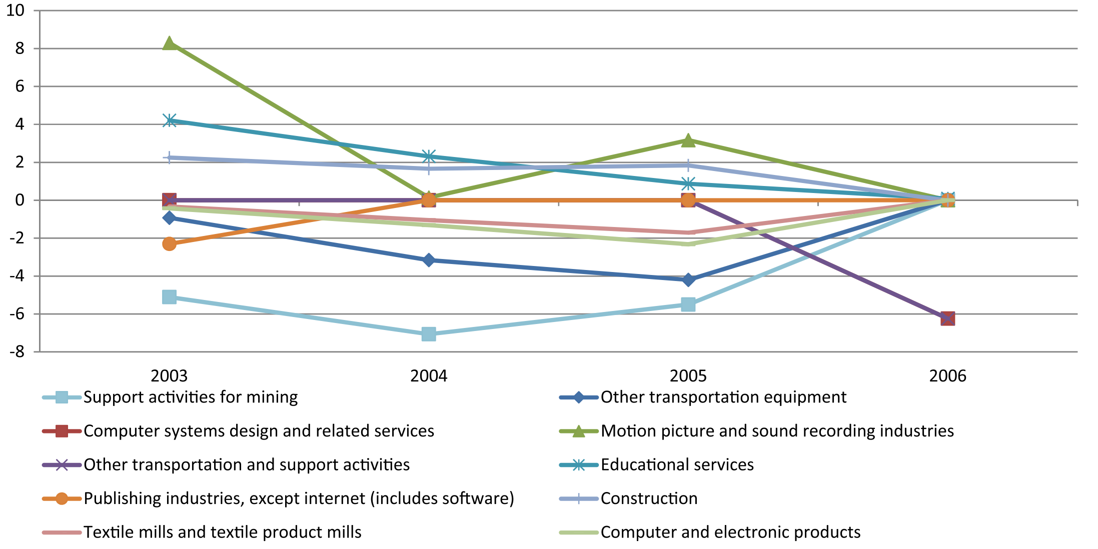

FIGURE 2 Discrepancy by industry (%) (three-digit NAICS) - 10 selected industries. NAICS = North American Industry Classification System

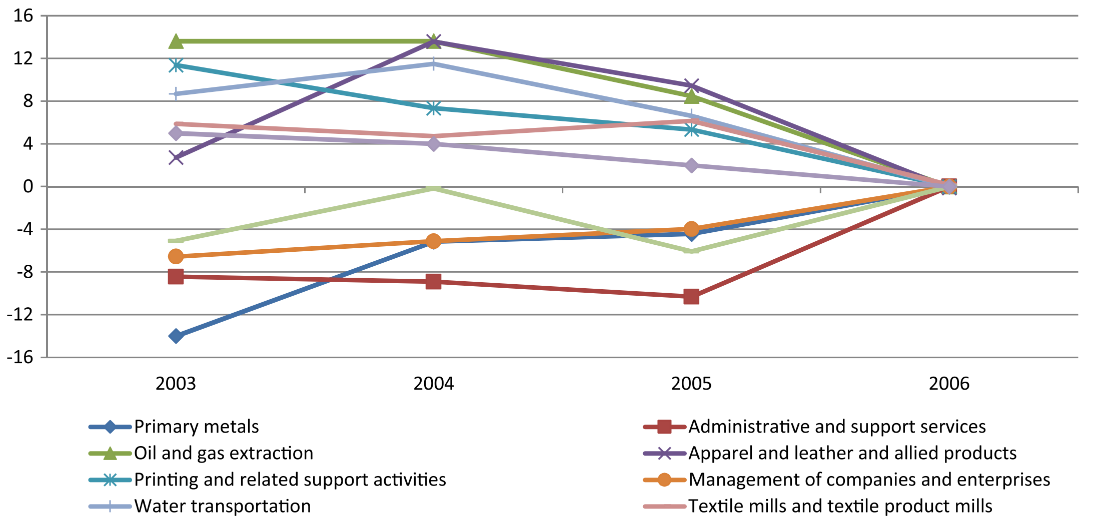

FIGURE 3 Discrepancy by commodity (%) (three-digit NAICS) - 10 selected commodities. NAICS = North American Industry Classification System

In addition to the vast number of series with zeroes in all periods, there are also many partially zero series. Most of these partially zero series are the result of changes between the 2002 and 2007 NAICS classification system. Some items in the 2002 benchmark IO tables ceased to exist in 2007, whereas some new items were introduced in the 2007 benchmark IO tables[2] that did not

[2]Some items in the 2002 benchmark IO tables (724 in the make table and 262 in the use table) have zero values for the newly introduced items; some items in the 2007 benchmarks IO tables (366 in the make table and 195 in the use table) have zero values because of the ceased-to-exist items. Twenty-nine in the use table and 10 in the make table have zero

TABLE 1 Series in the system of input–output accounts according to the three-digit NAICS and the six-digit NAICS level of detail

	Three-digit NAICS		Six-digit NAICS	
	No. of series	No. of nonzero series	No. of series	No. of nonzero series
Make table	4,355	802	707,160	9,069
Use table - Intermediate inputs	4,355	3,553	707,160	143,896
Use table - Final uses	737	294	387,660	4,221
Use table - Value added	195	193	2,490	2,406
Total	**9,642**	**4,842**	**1,804,470**	**159,592**

Note. NAICS = North American Industry Classification System.

previously exist. For newly introduced items, a small positive value was assigned (for computational reasons) to the 2002 benchmark. For items that ceased to exist in the 2007 table, the 2002 and annual values were typically very small and were removed prior to reconciliation. In addition, a small number of items have zero values in both benchmarks but have small values in the annual tables or vice versa. Items with zero values in both benchmarks but nonzero annual values were also removed. For items with nonzero benchmarks values but zero annual values, very small nonzero values were assigned (again for computational reasons) for the annual periods prior to reconciliation.

3 | THE RECONCILIATION OF A SYSTEM OF TIME SERIES

To impose accounting constraints in each component series of the annual accounts, the usual reconciliation procedures use a RAS balancing technique to enforce identities and reduce discrepancies as much as possible. In a recent study, Chen (2012) used a generalized least-squares procedure to reconcile GDP estimates from IO, final expenditures, and income accounts for a benchmark year according to the estimated reliabilities of initial source data items.

Requiring consistency in the time series of the national accounts system gives rise to a more general reconciliation problem, according to which temporal and contemporaneous constraints must be simultaneously satisfied (Di Fonzo, 1990; Di Fonzo & Marini, 2011, 2015).

The reconciliation problem can be formalized as follows. At the working level for this effort, the 852×830 make table matrix contains the gross output of 852 commodities from 830 industries. The use table consists of a 852×830 matrix of intermediate inputs, a 3×830 matrix of industry VA from industry income, and a 852×455 matrix of final uses. Let $\mathbf{X}_t, \mathbf{Z}_t, \mathbf{V}_t$, and $\mathbf{Y}_t$ denote the matrices of preliminary estimates of gross output, intermediate inputs, VA, and final uses, respectively, in the annual IO accounts for $t = 2003, \ldots, 2007$. Let $\bar{\mathbf{X}}_{2002}, \bar{\mathbf{Z}}_{2002}, \bar{\mathbf{V}}_{2002}$, and $\bar{\mathbf{Y}}_{2002}$ denote the corresponding matrices for benchmark year 2002, and $\bar{\mathbf{X}}_{2007}, \bar{\mathbf{Z}}_{2007}, \bar{\mathbf{V}}_{2007}$, and $\bar{\mathbf{Y}}_{2007}$ for benchmark year 2007.

The preliminary matrices can be conveniently rearranged into a one-dimensional vector of stacked time series.[3] Let $x_{t,i,j}$ denote the element (i,j) of the make table matrix $\mathbf{X}_t$, for $t =$

values in both benchmarks but have some small nonzero values in the annual tables, and 29 items in the use table are nonzero in both benchmarks but zero in the annual use tables.

[3]In order to link the reconciled series to the 2002 benchmarks, we consider the benchmark matrices of 2002 as part of the group of preliminary matrices as well.

2002, … , 2007 We consider all (i,j) elements of the matrices even if they are zeros for all or for some years. The $852 \cdot 830 = 707{,}160$ time series in the make table can be stacked into a single $4{,}242{,}960 \times 1$ vector as

$$\mathbf{x} = \begin{bmatrix} \mathbf{x}'_{2002} & \mathbf{x}'_{2003} & \mathbf{x}'_{2004} & \mathbf{x}'_{2005} & \mathbf{x}'_{2006} & \mathbf{x}'_{2007} \end{bmatrix}',$$

where each $\mathbf{x}_t = \text{vec}(\mathbf{X}_t)$, $t = 2002, \ldots, 2007$, is a column vector with 707,160 entries. Vectors $\mathbf{z}$, $\mathbf{y}$, $\mathbf{v}$ can also be set up in the same fashion. Their row dimensions are 4,242,960, 2,325,960, and 14,940, respectively. The input vector of preliminary data of the problem is thus defined as $\mathbf{p} = \begin{bmatrix} \mathbf{x}' & \mathbf{z}' & \mathbf{y}' & \mathbf{v}' \end{bmatrix}'$, where $\mathbf{p}$ has dimension $10{,}826{,}820 \times 1$. Let us now consider the constraints of the system. There are *exogenous* and *endogenous* constraints. The first type concerns the benchmark values for the years 2002 and 2007. Let $\mathbf{b}$ denote the vector of two-element time series from the benchmarked matrices previously defined, i.e.,

$$\mathbf{b} = \begin{bmatrix} \bar{x}_{2002,1,1} & \bar{x}_{2007,1,1} & \bar{x}_{2002,2,1} & \bar{x}_{2007,2,1} & \cdots & \bar{v}_{2002,3,829} & \bar{v}_{2007,3,829} & \bar{v}_{2002,3,830} & \bar{v}_{2007,3,830} \end{bmatrix}',$$

with dimension of $3{,}608{,}940 \times 1$. Let $\mathbf{H}_1$ denote the $3{,}608{,}940 \times 10{,}826{,}820$ mapping matrix for the exogenous constraints specified in $\mathbf{b}$ for the benchmark years 2002 and 2007. This matrix is given by

$$\mathbf{H}_1 = \begin{bmatrix} \mathbf{I}_K & \mathbf{0}_K & \mathbf{0}_K & \mathbf{0}_K & \mathbf{0}_K & \mathbf{0}_K \\ \mathbf{0}_K & \mathbf{0}_K & \mathbf{0}_K & \mathbf{0}_K & \mathbf{0}_K & \mathbf{I}_K \end{bmatrix},$$

where $\mathbf{I}_K$ and $\mathbf{0}_K$ are (square) identity and zero matrices, respectively, of order $K = 852 \cdot 830 + 852 \cdot 830 + 852 \cdot 455 + 3 \cdot 830 = 1{,}804{,}470$. Given that, as we have previously said, preliminary and benchmark 2007 values are different, it is $\mathbf{H}_1\mathbf{p} \neq \mathbf{b}$.

The endogenous constraints are defined by the set of accounting relationships defined by the IO tables. There are 852 row constraints (commodities) and 830 column constraints (industries) per year:

$$\sum_{j=1}^{830} x_{ij} - \sum_{j=1}^{830} z_{ij} - \sum_{f=1}^{455} y_{if} = 0 \qquad i = 1, \ldots, 852$$
$$\sum_{i=1}^{852} x_{ij} - \sum_{i=1}^{852} z_{ij} - \sum_{l=1}^{3} v_{lj} = 0 \qquad j = 1, \ldots, 830.$$

The aggregation constraint of total GDP equals total VA is redundant and can be disregarded, as it follows from adding up the first 1,682 constraints. In total, they add up to 10,092 constraints for $t = 2002, \ldots, 2007$. Let $\mathbf{H}_2$ denote the $10{,}092 \times 10{,}826{,}820$ matrix mapping the 10,826,820 elements in the preliminary vector to the 10,092 accounting constraints. Matrix $\mathbf{H}_2$ can be written as $\mathbf{H}_2 = \mathbf{I}_6 \otimes \mathbf{C}$, where

$$\mathbf{C} = \begin{bmatrix} \mathbf{1}'_{830} \otimes \mathbf{I}_{852} & -\left(\mathbf{1}'_{830} \otimes \mathbf{I}_{852}\right) & -\left(\mathbf{1}'_{455} \otimes \mathbf{I}_{852}\right) & \mathbf{0}_{852\times 2{,}490} \\ \mathbf{I}_{830} \otimes \mathbf{1}'_{852} & -\left(\mathbf{I}_{830} \otimes \mathbf{1}'_{852}\right) & \mathbf{0}_{830\times 387{,}660} & -\left(\mathbf{I}_{830} \otimes \mathbf{1}'_{3}\right) \end{bmatrix},$$

and $\mathbf{1}_k$ and $\mathbf{0}_{r\times c}$ denote a $(k \times 1)$ vector of ones and a $(r \times c)$ null matrix, respectively. Clearly, it is $\mathbf{H}_2\mathbf{p} \neq \mathbf{0}_{10{,}092\times 1}$.

In sum, we wish to derive the $10{,}826{,}820 \times 1$ vector of reconciled values $\mathbf{r}$ such that

$$\begin{bmatrix} \mathbf{H}_1 \\ \mathbf{H}_2 \end{bmatrix} \mathbf{r} = \begin{bmatrix} \mathbf{b} \\ \mathbf{0}_{10{,}092\times 1} \end{bmatrix} \tag{1}$$

and the differences in the temporal dynamics between $\mathbf{r}$ and $\mathbf{p}$ are minimized. It should be noted that the complete constraints matrix $\mathbf{H} = \begin{bmatrix} \mathbf{H}_1 \\ \mathbf{H}_2 \end{bmatrix}$ is a very large and sparse (i.e., many zero entries) matrix (the nonzero items are 22,921,680 out of 39,182,705,479,620, which means that about 99.9999415% of the entries is equal to zero), whose sparsity pattern can be exploited in the reconciliation process (Di Fonzo & Marini, 2011).

To reconcile a system of time series, we use adjustment procedures based on the linearly constrained optimization of two different quadratic objective functions:

- Proportional first difference adjustment (PFD):

$$\sum_{i=1}^{n}\sum_{t=2003}^{2007}\left(\frac{r_{t,i}}{p_{t,i}}-\frac{r_{t-1,i}}{p_{t-1,i}}\right)^2 \tag{2}$$

- Proportional adjustment (PROP):

$$\sum_{i=1}^{n}\sum_{t=2003}^{2007}\left(\frac{r_{t,i}-p_{t,i}}{p_{t,i}}\right)^2, \tag{3}$$

where PFD is a multivariate extension of the univariate benchmarking solution proposed by Denton (1971) and modified by Cholette (1984), and n is the number of nonnull variables of the system (4,842 and 159,592 for the NAICS 3 and NAICS 6 systems, respectively).

It should be noted that, unlike PFD, according to which the system is adjusted simultaneously (i.e., all variables and all years at the same time), the PROP criterion (3) can be simply achieved through the minimization of the five objective functions $\sum_{i=1}^{n}\left(\frac{r_{t,i}-p_{t,i}}{p_{t,i}}\right)^2$, $t = 2003, \dots, 2007$.

In fact, the adjustment principles operate very differently. The PROP criterion proportionally preserves the levels of the variables. On the other hand, the PFD criterion preserves the year-to-year movements of the variables. Because our target is to preserve the changes in the preliminary variables, we expect that the PFD method provides more satisfactory results for this exercise.

As we did in the first part of this research project (Chen et al., 2014), we also consider a combined objective function as follows:

$$\sum_{i\in S^{\mathrm{PFD}}}\sum_{t=2003}^{2007}\left(\frac{r_{t,i}}{p_{t,i}}-\frac{r_{t-1,i}}{p_{t-1,i}}\right)^2+\sum_{i\in S^{\mathrm{PROP}}}\sum_{t=2003}^{2007}\left(\frac{r_{t,i}-p_{t,i}}{p_{t,i}}\right)^2, \tag{4}$$

where both the PFD criterion and the PROP criterion are utilized (we call this criterion PFD–PROP). The variables in the system are divided in two subsets (S^{PFD} and S^{PROP}, respectively). The PFD criterion is used for those series showing meaningful and interpretable movements over time (namely, movements that we would like to preserve). Because PFD alone could have difficulty with series that have no meaningful movements between periods, to overcome this difficulty, the PFD–PROP procedure adjusts all these series according to PROP, while it maintains the PFD approach for the rest of the series .

It should be noticed that the combined objective function (4) improves and corrects the original proposal by Chen et al. (2013) and Chen et al. (2014), where the PROP part of the objective function was based on the improved normalized squared difference criterion (Huang, Kobayashi, & Tanji, 2008), which extends Friedlander (1961):

$$\sum_{i=1}^{n}\sum_{t=2003}^{2007}\frac{\left(r_{t,i}-p_{t,i}\right)^2}{\left|p_{t,i}\right|}. \tag{5}$$

The reconciliation procedure based on the constrained minimization of the combined objective function (4) is scale invariant (Bikker et al., 2013; see the Appendix), whereas using (5) in its PROP part is not. Scale invariance means that premultiplication of input data by a nonnegative constant λ produces reconciled values equal to those obtained using the original input data, multiplied by λ, which is a sensible property for a reconciliation procedure.[4]

For the constrained minimization problem, we use a direct solution of sparse and large systems implemented in MATLAB. The algorithm uses a multifrontal approach that solves the system through an LDL factorization (L and D denote a lower unit triangular matrix and a diagonal matrix, respectively) exploiting the sparsity of the system matrix. Using standard computing power, the code takes about 130 seconds to resolve the six-digit NAICS system using the simultaneous PFD solution. Further details about this algorithm were given in the work of Di Fonzo and Marini (2011).

4 | RESULTS

In this study, the revised (as described in Section 2) and not yet balanced annual IO tables from 2003 to 2007 (six-digit NAICS) are considered as preliminary series to be reconciled. In theory, the previously published annual IO tables should be used directly as the preliminary estimates. However, in the actual production, the previously published IO tables could not be directly used as the preliminary estimates because of the changes in the classification system and because new information from the benchmark revision of GDP needed to be incorporated.

In order to assess the global performance of the procedures, for each series ($i = 1, \dots, n$), we calculate the mean absolute adjustment (MAA) and the root mean squared adjustment (RMSA) to the percentage levels

$$\mathrm{MAA}_i^L = 100 \times \frac{1}{5} \sum_{t=2003}^{2007} \left| \frac{r_{t,i} - p_{t,i}}{p_{t,i}} \right| \qquad \mathrm{RMSA}_i^L = 100 \times \sqrt{\frac{1}{5} \sum_{t=2003}^{2007} \left(\frac{r_{t,i} - p_{t,i}}{p_{t,i}} \right)^2}$$

and to the percentage growth rates

$$\mathrm{MAA}_i^R = 100 \times \tfrac{1}{5} \sum_{t=2003}^{2007} \left| \frac{r_{t,i}}{r_{t-1,i}} - \frac{p_{t,i}}{p_{t-1,i}} \right| \qquad \mathrm{RMSA}_i^R = 100 \times \sqrt{\tfrac{1}{5} \sum_{t=2003}^{2007} \left(\frac{r_{t,i}}{r_{t-1,i}} - \frac{p_{t,i}}{p_{t-1,i}} \right)^2}.$$

To compare the results, we first consider all the variables of the system of accounts, and then, we focus on the 43 main aggregates of national accounts, which include GDP, gross output, intermediate inputs, and VA of 12 major industries, and six final expenditure categories. We compare the averages of indices calculated from all detailed reconciled series and from the 43 main aggregates using the three alternative procedures considered in Section 3. Table 2 shows the averages of indices MAA and RMSA calculated for the reconciled series in the IO tables at the most detailed level using the three alternative reconciliation procedures:

- Proportional first difference adjustment (PFD), based on criterion (2);
- Proportional adjustment (PROP), based on minimizing criterion (3);
- Combined PFD and PROP adjustment (PFD–PROP), as defined by criterion (4).

[4]We gratefully acknowledge one referee, who pointed out that the originally proposed formulation of the PFD–PROP procedure was not scale invariant and suggested us to change the combined objective function according to expression (4).

TABLE 2 Summary measures of adjustment of all detailed series

Criterion	Levels MAA^L	$RMSA^L$	Growth rates MAA^R	$RMSA^R$
PROP	3.398	6.701	8.687	1,397.877
PFD	0.740	6.356	1,471.108	1,303,600.325
PFD–PROP	0.334	3.574	2.723	1,385.103
PROP[a]	3.388	6.682	7.605	1,012.876
PFD[a]	0.153	1.401	0.335	53.105
PFD–PROP[a]	0.157	1.409	0.359	55.710

Note. MAA = mean absolute adjustment; RMSA = root mean squared adjustment; PROP = pure proportional; PFD = proportional first difference.
[a]Series with positive and negative values in the preliminary data are not considered.

TABLE 3 Summary measures of adjustment for 43 main aggregates

Criterion[a]	Levels MAA^L	$RMSA^L$	Growth rates MAA^R	$RMSA^R$
PROP	0.4401	0.9279	0.5599	1.0560
PFD	0.5431	1.0142	0.6400	1.0642
PFD–PROP	0.5807	1.1477	0.6819	1.1560
PROP-A	0.4516	0.9409	0.6816	1.1913
PFD-A	0.4967	1.1327	0.8079	1.6001
PFD–PROP-A	0.4971	1.1328	0.8091	1.6003

Note. MAA = mean absolute adjustment; RMSA = root mean squared adjustment; PROP = pure proportional; PFD = proportional first difference.
[a]A: From reconciliation at three-digit NAICS level of detail.

The summary measures in the top panel are computed from 159,592 reconciled series, whereas the summary measures in the bottom panel are computed from 158,379 reconciles series, excluding those with positive and negative values in the series. As it clearly appears, PFD–PROP offers the best results because it minimizes the adjustment to the full system of variables, with a pronounced gain over PFD, thanks to the different treatment of series with null values in the preliminary data. When these series are not taken into account (i.e., for 158,379 series out 159,592), the performances of PFD and PFD–PROP are very close. It should be noted that, when reconciliation was performed at three-digit NAICS level (Chen et al. 2014), we found a different result, namely, a pronounced gain of PFD–PROP over PFD.

Finally, to assess the impact of the level of detail on the results of the reconciliation, Table 3 shows the same indices for the main 43 national accounts aggregates from reconciled series at the detailed (six-digit NAICS) and aggregated (three-digit NAICS) levels of detail, using the three alternative procedures.

To calculate the summary measures for the 43 aggregates, preliminary and reconciled estimates at the six-digit NAICS level of detail (in the top panel) and those at the three-digit NAICS level of detail (in the bottom panel) were first aggregated to the 43 aggregates, (GDP, aggregates of gross output, intermediate inputs, and VA of 12 major industries, and aggregates of six major categories of final uses). The summary measures are computed based on the adjustment at the aggregated level.

We observe that, (i) for all the three alternative procedures, reconciliation at a more disaggregated level of detail produces smaller adjustment in percentage growth rates, and (ii) due

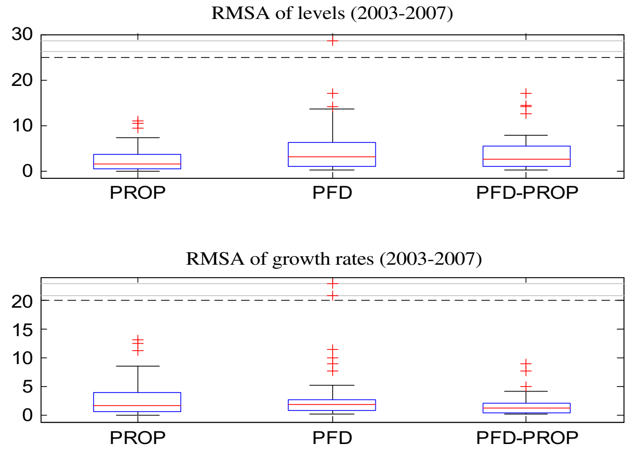

FIGURE 4 Boxplot of root mean squared adjustment (RMSA) statistics for the main 43 national accounts aggregates. PROP = pure proportional; PFD = Denton proportional first difference

to consistent compensations occurred when aggregating the component series before summary indices are computed, PROP turns out to be comparable with (and slightly better than) PFD and PFD–PROP.

Figure 4 displays the boxplots of RMSA_i^L (top chart) and RMSA_i^R (bottom chart) for the 43 aggregates (the absolute distance metric of MAA gives a less pronounced difference between the performance of the three procedures, and it is not shown). The visual inspection of the boxplots confirms that PFD–PROP produces the smallest adjustment of the growth rates, whereas PROP provides the best results in preserving the original levels. As for the growth rates, this conclusion is evident looking at the RMSA statistics.

To evaluate the adjustment produced by PFD when applied at different levels of detail, it is useful to look at the effect on some aggregate series, like GDP (Figure 5). The left-hand charts refer to the levels, the right-hand charts refer to the growth rates, and the adjustments to both levels and growth rates are shown in the bottom charts. In this case, the adjustments to GDP coming from the most detailed accounts are less pronounced than those produced by the series reconciled at three-digit NAICS (see Chen et al. 2014). This result comes true also for PROP and PFD–PROP, whose graphs are not shown to save space.

5 | USING ALTERABILITY COEFFICIENTS: A SENSITIVITY ANALYSIS

In a system of series, variables are often measured with different accuracy/reliability. It is thus rather natural that variables should be adjusted on the basis of their relative reliability (i.e., more accurate, less adjustment). Coefficients α_i, $i = 1, \dots, n$, can be used to manage the alterability of the n series in the system. These coefficients, which "artificially increase or reduce the covariance matrices of some series (…) so that these series are more or less affected by reconciliation" (Dagum and Cholette, 2006, p. 274), are usually restricted between 0 (unalterable) and 1 (maximum alterable).

The standard deviations of the preliminary estimates implicitly assumed by both PROP and PFD reconciliation procedures are

$$\sigma_{t,i} = \alpha_i \times |p_{t,i}|, \quad t = 2003, \dots, 2007, \quad i = 1, \dots, n.$$

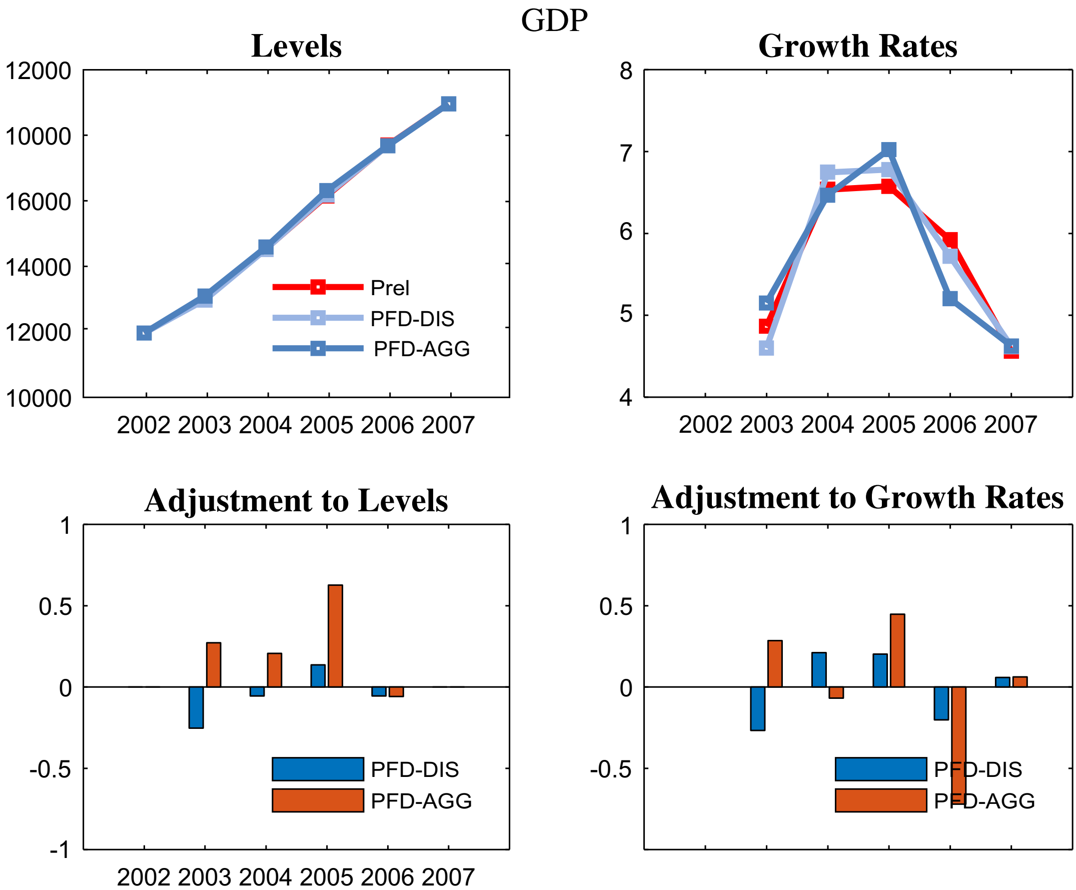

FIGURE 5 Gross domestic product (GDP) from disaggregated and aggregated reconciliation using Denton proportional first difference (PFD)

Thus, the modified PROP and PFD criteria are given by

- PROP with alterability coefficients:

$$\sum_{i=1}^{n}\frac{1}{\alpha_i^2}\sum_{t=2002}^{2007}\left(\frac{r_{t,i}-p_{t,i}}{p_{t,i}}\right)^2$$

- PFD with alterability coefficients:

$$\sum_{i=1}^{n}\frac{1}{\alpha_i^2}\sum_{t=2003}^{2007}\left(\frac{r_{t,i}}{p_{t,i}}-\frac{r_{t-1,i}}{p_{t-1,i}}\right)^2,$$

and the PFD–PROP criterion is defined accordingly. For practical purposes, in the following, the minimum value for α_i has been set equal to 0.001.

It should be noted that both PFD and PROP formulation *de facto* assume equal reliability of all the variables involved in the reconciliation.

We perform two distinct exercises to assess (i) the way the alterability coefficients can be used in our framework and (ii) the impact on the series subject to reconciliation, including the impact on the entire system as a whole and the distinctive impact on gross output, intermediate inputs, VA, and final uses time series.

TABLE 4 Aggregate and weighted MAA to the percentage growth rates as the alterability coefficient of output varies from 1% to 100%

			Alterability coefficient for output variables (%)										
Variable	**Matrix**	**n. series**	**1**	**10**	**20**	**30**	**40**	**50**	**60**	**70**	**80**	**90**	**100**
Gross output	**X**	9,069	0.03	0.08	0.15	0.20	0.23	0.25	0.25	0.25	0.25	0.24	0.24
Intermediate inputs	**Z**	143,896	0.54	0.58	0.61	0.60	0.58	0.56	0.54	0.51	0.50	0.48	0.47
Value added	**V**	2,406	0.50	0.44	0.33	0.24	0.19	0.16	0.14	0.12	0.12	0.11	0.11
Final uses	**Y**	4,221	0.37	0.31	0.20	0.14	0.09	0.09	0.11	0.12	0.13	0.14	0.14
Weighted MAA^R			0.51	0.54	0.57	0.56	0.54	0.52	0.50	0.48	0.47	0.45	0.44

Note. MAA = mean absolute adjustment.

In the first exercise, reconciled estimates are obtained by varying alterability coefficients from 0.01 to 1.0 in increments of .10 for the 9,069 nonzero time series of the gross output matrix, leaving the other time series free to vary, that is, alterability coefficients for intermediate inputs, VA, and final uses are assigned to 1.0. This exercise aims at testing the use of alterability coefficients to control the adjustment of a subset of variables in the system (in this case, the subset contains all component series of the gross output matrix). Smaller adjustments should be made to variables with smaller values of the alterability coefficient.

Assessment of the impact of the reconciliation on a single time series is made using the MAA to the percentage growth rates (MAA^R), as defined in the previous section. Table 4 presents the averages of MAA_i^R calculated for the aggregates pertaining to matrices **X** (Gross Output), **Z** (Intermediate consumption on inputs), **V** (VA), and **Y** (final uses), respectively. Weighted MAA^R is the aggregated MAA_i^R indices weighted by the number of series in each matrix as a percentage of the total number of series in the system. According to this metric, the minimum adjustment is achieved when all variables have the same alterability (i.e., a pure PFD adjustment, as shown in the last column).

Figure 6 depicts the adjustment paths of gross output, intermediate inputs, VA, and final uses as the alterability coefficient for gross output increases from 1% to 100%. MAA^R of gross output rises from .03 to .25 percentage points (p.p.) as its alterability coefficient increases from 1% to 50% and, then, stays stable thereafter. The increase in the alterability coefficient for gross output initially led to an increase in the adjustment in intermediate inputs. However, the increase in MAA^R reaches its peak as the alterability coefficient for gross output reaches 20%; it starts to decline as the alterability coefficient for gross output continues to increase. In contrast, the rise in the adjustment in gross output as its alterability coefficient increases is responded with considerable decreases in the adjustment of VA and final uses; as the adjustment of gross output stabilizes, the adjustment of these two variables also stabilizes.

In the second exercise, we perform a sensitivity analysis on gross operating surplus (GOS). We replicate a typical balancing exercise of national accounts tables, where the alterability coefficients of the more reliable variables in the system are assigned to 0.01. These variables are gross output, final uses, and the two components of VA (wages and salaries and net taxes). Intermediate inputs are given an alterability coefficient of 1.0 because it is considered less reliable than other variables. As a baseline of this exercise, we assign the alterability coefficient for GOS to 0.01 and balance the system of the accounts. We then let the alterability coefficient for GOS increase from 0.01 to 0.02 in increments of 0.001 and to increase it from 0.1 to 0.3 in increments of 0.1. We measure how the adjustments to GOS and to other variables change as the alterability coefficient for GOS increases.

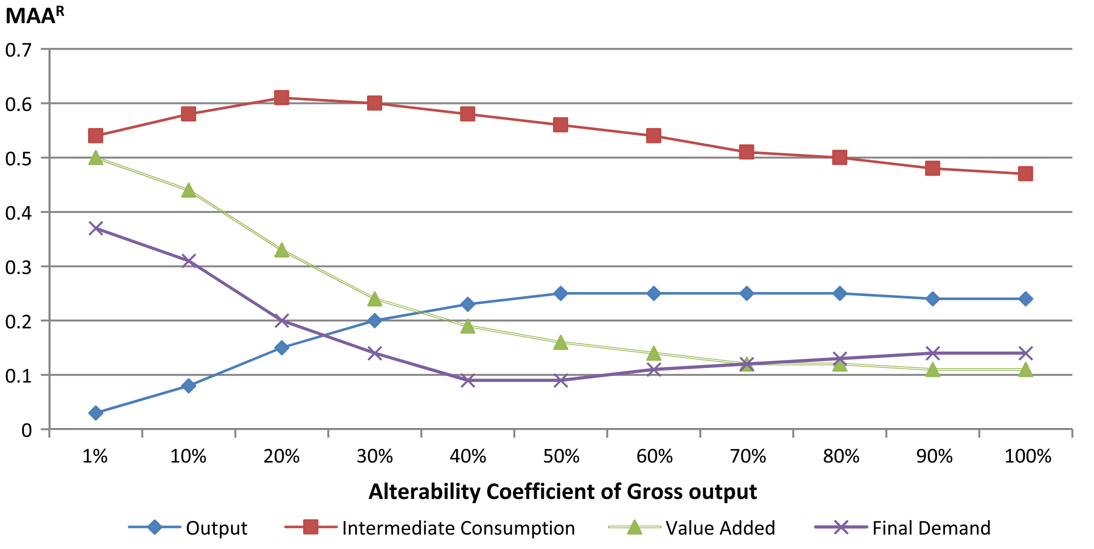

FIGURE 6 Mean absolute adjustment to the percentage growth rates (MAA^R) of variables in the input–output accounts as the alterability coefficient of gross output varies from 1% to 100%

TABLE 5 Mean absolute adjustment to the percentage growth rates as the alterability coefficient of gross operating surplus varies from 1% to 30%

	Alterability coefficient for gross operating surplus (%)													
Variable	**1**	**1.1**	**1.2**	**1.3**	**1.4**	**1.5**	**1.6**	**1.7**	**1.8**	**1.9**	**2**	**10**	**20**	**30**
Gross operating surplus	*0.10*	*0.11*	*0.12*	*0.14*	*0.15*	*0.16*	*0.17*	*0.18*	*0.19*	*0.20*	*0.21*	*0.36*	*0.39*	*0.40*
Gross output	0.07	0.07	0.07	0.07	0.06	0.06	0.06	0.06	0.06	0.06	0.06	0.06	0.06	0.06
Intermediate inputs	0.11	0.11	0.11	0.11	0.11	0.11	0.11	0.11	0.11	0.11	0.11	0.11	0.11	0.11
GDP	0.07	0.08	0.08	0.08	0.09	0.09	0.09	0.10	0.10	0.10	0.10	0.15	0.15	0.15
Wages and salaries	0.06	0.06	0.06	0.06	0.06	0.05	0.05	0.05	0.05	0.05	0.05	0.02	0.02	0.02
Net taxes	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.01	0.01	0.01	0.00	0.00	0.00
Personal consumption	0.07	0.07	0.07	0.06	0.06	0.06	0.05	0.05	0.05	0.05	0.05	0.03	0.03	0.03
Private fixed investment	0.26	0.26	0.26	0.26	0.26	0.26	0.26	0.26	0.26	0.26	0.26	0.25	0.25	0.25
Inventories	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01
Exports	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.03	0.03
Imports	0.36	0.36	0.36	0.36	0.36	0.36	0.36	0.36	0.36	0.36	0.36	0.37	0.37	0.37
Govern., cons., and inv.	0.08	0.08	0.08	0.07	0.07	0.07	0.07	0.07	0.07	0.07	0.07	0.06	0.05	0.05

Note. GDP = gross domestic product.

Table 5 presents the averages of MAA_i^R calculated for the aggregates pertaining to the variables in the IO accounts and the major components of GDP. The adjustment to GOS rises from 0.10 to 0.21 p.p. as its alterability coefficient increases from 1% to 2%. Note that MAA^R for GOS grows almost linearly with the alterability coefficient in this range. As its alterability coefficient continues to increase from 2% to 30%, the MAA^R for GOS reaches 0.40 p.p. and remains stable thereafter.

Figure 7 displays the impact of varying alterability coefficient of GOS on the variables in the IO accounts. Corresponding to the rise in the adjustment in GOS from 0.10 to 0.40 p.p. as its alterability coefficient increases from 1% to 30%, the adjustments in gross output, wages and salaries,

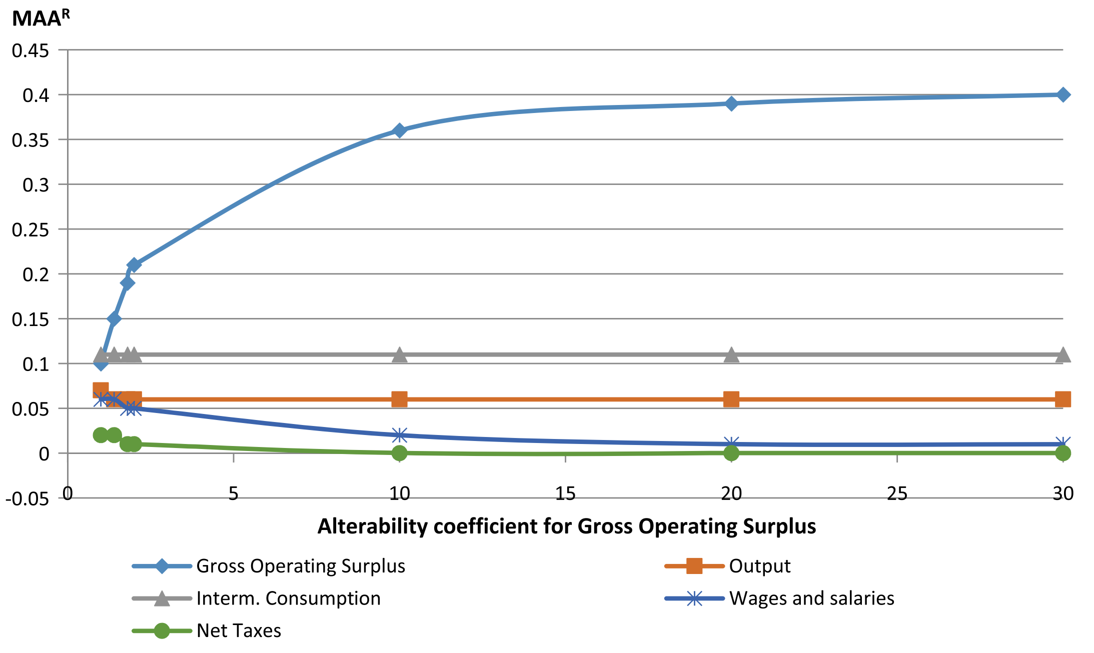

FIGURE 7 Mean absolute adjustment to the percentage growth rates (MAA^R) of variables in the input–output accounts as the alterability coefficient of gross operating surplus (GOS) varies from 1% to 30%

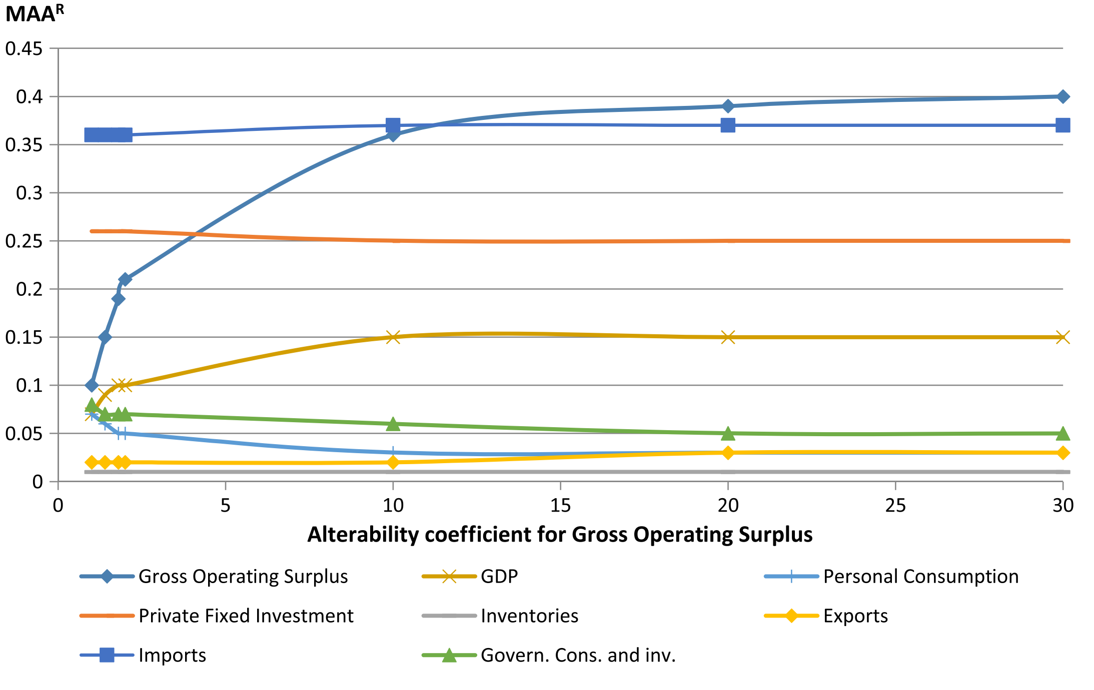

FIGURE 8 Mean absolute adjustment to the percentage growth rates (MAA^R) of major components of gross domestic product (GDP) as the alterability coefficient of gross operating surplus (GOS) varies from 1% to 30%

and net taxes are affected only mildly initially, whereas the adjustment in intermediate inputs is not affected at all. MAA^R in all variables remains stable as the alterability coefficient of GOS reaches 20%. MAA^R of GOS reaches a maximum value of 0.41 at a 50% level and remains such for higher values of the alterability coefficient.

Similarly, Figure 8 shows that changes in the alterability coefficient for GOS have little or no impact on the adjustment in exports, imports, inventory change, and private fixed investment. Adjustment in government consumption and investment and personal consumption declines mildly as the alterability coefficient for GOS increases from 1. To absorb the increase in the adjustment of GOS from 0.10 to 0.36 p.p. as its alterability coefficient increases from 1% to 10%, the adjustment in GDP increases from 0.10 to 0.15 p.p. The adjustment in GDP is not further impacted as the alterability coefficient for GOS increases from 10% to 30%.

We have two observations from both exercises. First, varying the alterability coefficient of one variable has different degrees of impact on the adjustment of variables in the system. The adjustment of some variables may barely be impacted. Second, increases in the alterability coefficient of one variable do not have continued impact on the adjustment of the variable itself or on the adjustment of other variables in the system. Changes in adjustment of each variable eventually stabilize as the alterability coefficient reaches a certain level.

6 | CONCLUSIONS

In this paper, we have demonstrated that a simultaneous least-squares procedure based on a movement preservation principle is a sensible and feasible approach for reconciling large disaggregated time series of accounts with quinquennial benchmarks available from IO tables. Our objective was to minimize the impact of the adjustment on the movements in the preliminary series. In general, we have found that this objective is best achieved through a constrained optimization procedure based on a movement preservation principle, that is, in our case, the PFD criterion proposed by Denton (1971) and modified by Cholette (1984). Looking at the temporal dynamics of the data, the PFD-based procedure is able to smooth the differences observed between the preliminary and the benchmark data of 2007, reducing the impact of the correction by distributing it over all the years.

However, we have noticed that the adoption of a pure PFD adjustment provides unsatisfactory results for very small series with no meaningful movements and series with positive and negative values. Because these movements are more difficult to preserve, these series should be adjusted according to a simple proportional criterion. We have shown that a constrained optimization procedure that minimizes a combined PFD–PROP objective function improves the overall adjustment of the system, minimizing the impact on the year-to-year changes of the preliminary series. We have also shown reconciliation at a disaggregated level always minimizes total adjustments than reconciliation at a more aggregated level. Moreover, we have demonstrated the flexibility of the constrained optimization procedure to limit the adjustment of specific variables in reconciliation via alterability coefficients according to production requirements.

DISCLAIMER

The views expressed herein are those of the authors, do not represent the official policies of the Bureau of Economic Analysis, and should not be attributed to the IMF, its Executive Board, or its management.

ORCID

Tommaso Di Fonzo http://orcid.org/0000-0003-3388-7827

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APPENDIX

SCALE INVARIANCE OF THE PFD–PROP RECONCILIATION PROCEDURE

According to Bikker et al. (2013), for a reconciliation procedure to be scale invariant, it is needed that, if all input data are multiplied by a nonnegative constant λ, the new reconciled values should be equal to the reconciled values obtained using the original input data, multiplied by the constant λ.

In formal terms, calling $\mathbf{p}$ and $\mathbf{g}$ the original preliminary and benchmark data, respectively, and considering $\mathbf{p}^* = \lambda\mathbf{p}$ and $\mathbf{g}^* = \lambda\mathbf{g}$, if we denote $\mathbf{r} = \mathcal{R}(\mathbf{p}, \mathbf{g})$ and $\mathbf{r}^* = \mathcal{R}(\mathbf{p}^*, \mathbf{g}^*)$ the reconciled estimates produced by the reconciliation procedure $\mathcal{R}$ working on the original and transformed input data, respectively, the reconciliation procedure $\mathcal{R}$ is scale invariant if $\mathbf{r}^* = \lambda\mathbf{r}$.

Now, without loss of generality, let us consider the n nonnull variables of the system as $n = n_1 + n_2$ variables, the former n_1 treated according to the PFD criterion (2), and the latter according to the PROP criterion (3). Assuming that no item in $\mathbf{p}$ is equal to zero, let us split the $(m \times 1)$ vector of preliminary values $\mathbf{p}$ as $\mathbf{p} = \begin{bmatrix}\mathbf{p}_1' & \mathbf{p}_2'\end{bmatrix}'$, where $\mathbf{p}_1 \in S^{\text{PFD}}$ is the $(m_1 \times 1)$ vector of preliminary values managed through PFD and $\mathbf{p}_2 \in S^{\text{PROP}}$ is the $(m_2 \times 1)$ vector of preliminary values managed through PROP. It is worth noting that, for the problem in hand, the n nonnull variables of the system cover six years, that is, $m = 6n$, $m_1 = 6n_1$, and $m_2 = 6n_2$. The $(m \times 1)$ reconciled vector $\mathbf{r} = \begin{bmatrix}\mathbf{r}_1' & \mathbf{r}_2'\end{bmatrix}'$ is defined accordingly.

The combined objective function (4) can be written in matrix form as

$$(\mathbf{r} - \mathbf{p})'\Omega(\mathbf{r} - \mathbf{p}) = (\mathbf{r}_1 - \mathbf{p}_1)'\Omega_1(\mathbf{r}_1 - \mathbf{p}_1) + (\mathbf{r}_2 - \mathbf{p}_2)'\Omega_2(\mathbf{r}_2 - \mathbf{p}_2),$$

where

$$\Omega = \begin{bmatrix}\Omega_1 & \mathbf{0} \\ \mathbf{0} & \Omega_2\end{bmatrix}$$

is a block-diagonal matrix, with $\Omega_1 = \hat{\mathbf{p}}_1^{-1}\Delta'\Delta\hat{\mathbf{p}}_1^{-1}$ and $\Omega_2 = \hat{\mathbf{p}}_2^{-1}\hat{\mathbf{p}}_2^{-1}$; $\hat{\mathbf{p}}_1 = \text{diag}(\mathbf{p}_1)$ and $\hat{\mathbf{p}}_2 = \text{diag}(\mathbf{p}_2)$ are $(n_1 \times n_1)$ and $(n_2 \times n_2)$ diagonal matrices, respectively; and

$$\Delta = \mathbf{I}_{n_1} \otimes \begin{bmatrix} -1 & 1 & 0 & 0 & 0 & 0 \\ 0 & -1 & 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 1 & 0 & 0 \\ 0 & 0 & 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 & -1 & 1 \end{bmatrix}$$

is the exact first differences matrix working on all the n_1 variables belonging to S^{PFD}.

The PFD–PROP reconciled estimates obtained from the original input data $\mathbf{p}$ and $\mathbf{g}$ are the solution to the following linearly constrained quadratic problem:

$$\mathbf{r}_{\text{PFD-PROP}} = \arg\min_{\mathbf{r}} (\mathbf{r}-\mathbf{p})'\Omega(\mathbf{r}-\mathbf{p}) \quad \text{s.t.} \quad \mathbf{Hr}=\mathbf{g}.$$

Similarly, the PFD–PROP reconciled estimates for the transformed input data $\mathbf{p}^*$ and $\mathbf{g}^*$ are the solution to the constrained problem:

$$\mathbf{r}^*_{\text{PFD-PROP}} = \arg\min_{\mathbf{r}^*} (\mathbf{r}^*-\mathbf{p}^*)'\Omega^*(\mathbf{r}^*-\mathbf{p}^*) \quad \text{s.t.} \quad \mathbf{Hr}^*=\mathbf{g}^*, \tag{A1}$$

where

$$\Omega^* = \begin{bmatrix} \frac{1}{\lambda^2}\Omega_1 & \mathbf{0} \\ \mathbf{0} & \frac{1}{\lambda^2}\Omega_2 \end{bmatrix} = \frac{1}{\lambda^2}\Omega.$$

If we call $\tilde{\mathbf{r}} = \dfrac{\mathbf{r}^*}{\lambda}$, after simple calculation, the minimand and the constraint of expression (A1) become, respectively,

$$(\tilde{\mathbf{r}}-\mathbf{p})'\Omega(\tilde{\mathbf{r}}-\mathbf{p}) \quad \text{and} \quad \mathbf{H}\tilde{\mathbf{r}}=\mathbf{g}.$$

Thus, $\mathbf{r}_{\text{PFD-PROP}} = \tilde{\mathbf{r}}_{\text{PFD-PROP}}$, from which it follows the result $\mathbf{r}^*_{\text{PFD-PROP}} = \lambda\mathbf{r}_{\text{PFD-PROP}}$, showing the scale invariance of the PFD–PROP reconciliation procedure.